# AL-KAWTHAR UNIVERSITY

### PROGRAMMING FUNDAMENTALS PROJECT DOCUMENTATION

### BANK MANAGEMENT SYSTEM

**Submitted To: Ms. Nida-e-Rub**


**Submitted By**

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**Degree Program**

**BS Computer Science (CS)**

**Project Type**

**Console based Application in C Language**

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## 1. Project Overview

The Bank Management System is a CLO-based application developed using the C programming language. The project is designed to help manage customer bank accounts and perform daily banking operations efficiently. It provides separate functionalities for Admin, Bank Employees, and Customers while maintaining customer records and transaction details.

The system includes account creation, deposits, withdrawals, balance inquiry, account searching, updating customer information, deleting accounts, viewing records, verification processing, and complaint management.

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## 2. Introduction

Banking institutions handle a large amount of customer information and financial transactions every day. Managing these records manually is time-consuming and increases the possibility of errors.

The Bank Management System automates these activities through a CLO application, making customer management faster, more organized, and easier to maintain. The project also demonstrates the practical implementation of programming concepts learned in Programming Fundamentals.

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## 3. Problem Statement

Manual banking systems require significant paperwork and human effort, making data management inefficient and error-prone. There is a need for a computerized solution that allows secure account management, transaction processing, and quick access to customer information.

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## 4. Objectives

- Develop a CLO-based Bank Management System using C language.

- Manage customer account information efficiently.
- Allow secure account creation and maintenance.
- Implement deposit and withdrawal operations.
- Enable balance inquiry and account searching.
- Update and delete customer records.
- Provide separate functionalities for Admin, Employee, and Customer.
- Practice programming concepts including functions, arrays, conditional statements, loops, and file handling.

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## 5. Scope of the Project

The project focuses on providing basic banking functionalities in an educational environment. It allows management of customer records and financial transactions while demonstrating CLO development and modular programming in C.

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## 6. Users of the System

**Admin**

- Login to the system
- Add employee accounts
- Delete emloyee accounts
- View all accounts
- Search accounts
- Edit and update records
- Logout

**Bank Employee**

- Login to the system
- Add accounts
- Delete accounts

- Search customer records
- Update account information
- Verify customer requests
- View complaints
- Logout

**Customer**

- Sign up
- Login
- Submit account creation request
- Check verification status
- Reapply if application is rejected
- Deposit money
- Withdraw money
- Check account balance
- View transaction history
- File complaints and check status
- Logout

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## 7. Functional Requirements

The Bank Management System provides the following functionalities:

- Create new bank accounts
- Store customer details
- Deposit money
- Withdraw money
- Check account balance
- Search account using ID or name
- Update customer information

- Delete accounts
- Display all account records
- User-friendly menu navigation
- Verification processing
- Complaint handling
- Transaction history viewing
- Employee can reply to complaints
- Minimum balance of 500 is required after withdrawal
- Login has 3 attempts for security

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## 8. System Architecture

The system is divided into three major modules:

**Admin Module**

Responsible for managing employee accounts and administrative operations.

**Bank Employee Module**

Handles account management related to customers, verification requests, and complaint monitoring.

**Customer Module**

Allows customers to register, login, manage transactions, and submit complaints.

All modules communicate with the central banking records and provide authorized access according to user roles.

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## 9. Modules Description

**Account Management Module**

Creates and stores customer accounts including personal information and account balance.

**Transaction Module**

Handles deposits, withdrawals, and balance inquiries while updating records.

**Search Module**

Searches customer accounts using account ID or customer name.

**Update Module**

Allows authorized users to modify customer information.

**Delete Module**

Removes customer accounts from the system when required.

**Complaint Module**

Allows customers to submit complaints which can be viewed by bank employees.

**Verification Module**

Processes customer account verification requests and displays verification status.

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### 10. Development Progress Log

| Date | Task Completed | Meeting Mode | Remarks |
| --- | --- | --- | --- |
| **09-Apr-2026** | Initial project planning and requirement gathering | In University | Finalized project idea and identified Admin, Employee, and Customer modules. |
| **12-Apr-2026** | Prepared project proposal | Zoom | Defined project scope and Console approach. |
| **18-Apr-2026** | Designed login screens and menus | WhatsApp | Created navigation structure for all users. |
| **24-Apr-2026** | Developed account creation module | In University | Implemented customer data entry and account storage. |
| **30-Apr-2026** | Implemented deposit functionality | WhatsApp | Added amount validation and balance updates. |

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|---|---|---|---|
| **05-May-2026** | Implemented withdrawal functionality | Zoom | Ensured withdrawal only with sufficient balance. |
| **12-May-2026** | Added balance inquiry and search | In University | Enabled searching by account ID and customer name. |
| **20-May-2026** | Developed update and delete features | WhatsApp | Allowed modification and removal of records. |
| **28-May-2026** | Added customer verification and complaint handling | Zoom | Integrated verification workflow and complaint section. |
| **05-Jun-2026** | Integrated all modules | In University | Connected Admin, Employee, and Customer functionalities. |
| **12-Jun-2026** | Testing and debugging | WhatsApp | Fixed logical errors and improved reliability. |
| **18-Jun-2026** | Documentation and final review | Online | Prepared final report and verified project features. |

# 11. Testing

The project was tested using different scenarios:

- New account creation
- Valid login
- Deposit operation
- Withdrawal operation
- Balance inquiry
- Search functionality
- Update records

- Delete records
- Verification status
- Complaint submission

Testing confirmed that the implemented modules performed according to the intended functionality.

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## 12. Current Limitations

- Designed for educational purposes.
- Security features can be enhanced.
- Data is stored temporarily using arrays
- Data will be lost when program closes
- Additional validation checks can be implemented.

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## 13. Future Improvements

- Online fund transfer functionality
- SMS and email notifications
- Advanced security features
- Database integration using SQL
- ATM simulation
- Loan management system
- Mobile banking support

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## 14. Technologies Used

**Programming Language:** C

**Application Type:** Console-Based

**Programming Concepts Used:**

- Functions
- Arrays
- Conditional Statements
- Loops
- String Manipulation

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## 15. Conclusion

The Bank Management System successfully demonstrates the implementation of a CLO-based banking application using the C programming language. The project provides an organized approach to managing customer accounts and banking transactions through Admin, Bank Employee, and Customer modules.

It helped strengthen understanding of programming fundamentals, modular design, user interaction, and practical software development. The system serves as a solid foundation for future enhancements and real-world banking applications.